B.Sc. Engg. Part I, Even Semester Examination 2022
Dept. of Computer Science and Engineering
Course Code: PHY 1211
Course Title: Basic Electricity and Electrical Circuits

1. a) State Gauss' theorem and apply it to obtain an expression for the electric field at a point
i) Outside and
ii) Inside a charged conducting sphere.

Solve :

(a) *Field for points outside the sphere ($r > R$)*

Let us consider a point P outside the sphere at a distance r from the centre of the sphere. Consider a spherical Gaussian surface of radius r passing through the point. From symmetry and uniformity of the density, the field is radial and uniform over the Gaussian surface. Hence the angle between E and dS will be zero. Applying Gauss' law, we have

$$\epsilon_0 \oint E.dS = \epsilon_0 \oint E\cos 0° dS$$

$$= \epsilon_0 E4\pi r^2 = \int \rho dV \qquad (2.12)$$

Since there is no charge outside the spherical region of charges, *i.e.*, $\rho = 0$ for $r > R$, the right side of eqn. 2.12 becomes

$$\int \rho dV = \rho . \frac{4}{3}\pi R^3$$

Hence eqn. 2.12 gives

$$E.4\pi r^2 = \frac{1}{\epsilon_0} \rho . \frac{4\pi}{3} R^3$$

$$\text{or, } E = \frac{\rho}{3\epsilon_0} \frac{R^3}{r^2} \quad \text{for } r > R.$$

If q be the total charge within the spherical region, then

$$\int \rho dV = q$$

Hence eqn. 2.12 reduces to

$$\epsilon_0 E . 4\pi r^2 = q$$

$$\text{or, } E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \text{for } r > R. \qquad (2.13)$$

Thus for $r > R$, the total distribution of charges may be considered to be concentrated at the centre of the spherical region.

(b) *E for points inside the sphere ($r < R$)*

Now consider a point P at a distance r from the centre of the sphere of charge. Imagine a spherical Gaussian surface of radius r passing through the point. Again from symmetry consideration, E and dS will point in the same direction. Applying Gauss' law we obtain

$$\epsilon_0 \oint E . dS = \epsilon_0 \oint E\cos 0° dS$$

$$= \epsilon_0 E \oint dS = \epsilon_0 E . 4\pi r^2 = \int \rho dV = q' \text{ (say)}$$

where q' is the portion of q contained within the sphere of radius r.

$$\therefore \quad E = \frac{1}{\epsilon_0} \frac{q'}{4\pi r^2} = \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2}$$

Now $q = \rho . \frac{4}{3}\pi R^3$ and $q' = \rho . \frac{4}{3}\pi r^3$

$$\therefore \quad q' = q \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = q \left(\frac{r}{R}\right)^3$$

Hence the expression for E for points inside the sphere ($r < R$) becomes

$$E = \frac{1}{4\pi\epsilon_0} \frac{qr}{R^3} \quad \text{for} \quad r < R. \qquad (2.14$$

b) What is an electric dipole? Calculate the Electric field due to an electric dipole at a point on the perpendicular bisector of the dipole.

An electric dipole consists of two equal but opposite charges separated by a small distance. Fig. 1.8 shows an electric dipole. The charges are +q and −q and they are separated by a distance 2a.

We would like to calculate the field **E** at a point P due to the dipole. P is located at a distance x along the perpendicular bisector of the line joining the charges. We shall assume that x >> a.

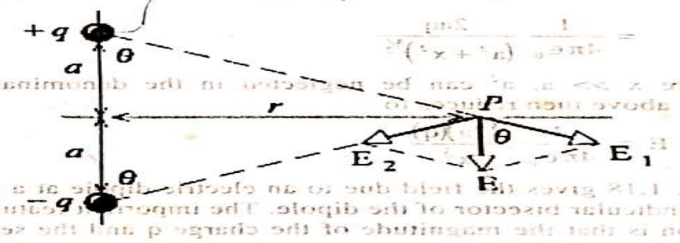


Fig. 1.8

Let the distance of the point P be y from both the charges. Let $\mathbf{E}_1$ be the electric field at the point P due to the charge +q and $\mathbf{E}_2$ be that due to −q. The total field at P due to the dipole is obtained by vector addition of

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$$

Since the charges have the same magnitude and the distance of the point P from the charges is also same, the magnitudes of the fields E_1 and E_2 are equal. Or,

$$E_1 = E_2 = \frac{1}{4\pi\epsilon_0}\frac{q}{y^2} = \frac{1}{4\pi\epsilon_0}\frac{q}{a^2+x^2}$$

The directions of $\mathbf{E}_1$ and $\mathbf{E}_2$ are indicated by arrows in Fig. 1.8. The horizontal components of $\mathbf{E}_1$ and $\mathbf{E}_2$ cancel each other. Hence the vector sum of $\mathbf{E}_1$ and $\mathbf{E}_2$ points vertically downwards and has the magnitude

$$E = 2E_1 \cos\theta$$

From the figure

$$\cos\theta = \frac{a}{y} = \frac{a}{\sqrt{a^2+x^2}}$$

Substituting the values of E_1 and $\cos\theta$ in eqn. 1.17 we obtain

$$E = \frac{2}{4\pi\epsilon_0}\frac{q}{(a^2+x^2)}\cdot\frac{a}{\sqrt{a^2+x^2}}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{2aq}{(a^2+x^2)^{3/2}}$$

since x >> a, a^2 can be neglected in the denominator. The equation above then reduces to

$$E \equiv \frac{1}{4\pi\epsilon_0}\cdot\frac{(2a)(q)}{x^3} \qquad (1.18)$$

c) Two parallel plates of area 20c * m ^ 2 each are separated by a distance of 10 mm. If one plate carries a charge 2 * 10 ^ - 9 * C and the other carries a charge - 2 * 10 ^ - 9 * C calculate the eclectic field in the space between them.

***Example 2.6** Two parallel plates of area 20 cm^2 each are separated by a distance of 10 mm. If one plate carries a charge $+2 \times 10^{-9}$C, and the other carries a charge -2×10^{9}C, calculate the electric field in the space between them.*

Soln.

Surface charge density for each of the plate is given by

$$\sigma = \frac{2\times10^{-9}}{20\times10^{-4}}\text{Cm}^{-2} = 10^{-6}\ \text{Cm}^{-2}.$$

Since one of the plate carries positive charge while the other carries negative charge, the electric field in the space between them

$$E = \frac{\sigma}{\epsilon_0} = \frac{10^{-6}\text{Cm}^{-2}}{8.85\times10^{-12}\text{C}^2\text{N}^{-1}\text{m}^{-2}} = 1.13 \times 10^5\ \text{N/C}.$$

2. (a) Define

(i) Susceptibility

(ii) Permeability

(iii) Self Inductance

Solve :

i) **Susceptibility** measures how easily a material can be magnetized when placed in an external magnetic field. It's a dimensionless factor that signifies the degree of magnetization induced in the material compared to the applied field.

ii) **Permeability** describes how easily a material allows magnetic fields to pass through it. It's a measure of the material's ability to concentrate magnetic flux compared to free space. Permeability is measured in Henrys per meter (H/m) and has a constant value for free space (μ_0).

iii) **Self-inductance** is the property of a conductor to oppose changes in the current flowing through it by inducing an electromotive force (EMF) within itself. This induced EMF is proportional to the rate of change of the current. Self-inductance is measured in Henrys (H) and depends on the geometry and material of the conductor.

b) Two coils are wound on the same iron rod so that the flux generated by one passes through the other also. The primary coils has N_{p} loops on it and when a current of 2A flows through it, the flux in it is 2.5 * 10 ^ - 4 * wb Determine the mutual inductance of the two coils if the secondary coils has N_{s} loops on it.

Example 9.14 *Two coils are wound on the same iron rod so that the flux generated by one passes through the other also. The primary coil has N_p loops on it and when a current of 2A flows through it, the flux in it is 2.5×10^{-4} Wb. Determine the mutual inductance of the two coils if the secondary coil has N_S loops on it.*

Soln.

Let us assume that the secondary coil is an open circuit. Now

$$\mathcal{E}_s = N_s \frac{\Delta\phi_S}{\Delta t}; \qquad \text{also} \qquad \mathcal{E}_s = M \frac{\Delta i_p}{\Delta t}$$

From the two equations we get

$$M = N_s \frac{\Delta\phi_S}{\Delta i_p}$$

Now $\Delta\phi_S = (2.5 \times 10^{-4} - 0)$ Wb

and $\Delta i_p = (2 - 0)$A

$$\therefore \quad M = N_s \frac{(2.5\times10^{-4} - 0)\text{Wb}}{(2-0)\text{A}} = (1.25 \times 10^{-4} N_s)\text{H}.$$

c) Describe an expression for Gauss' law as applied to a dielectric.

Di-electric and gauss's Law:

Let us consider two parallel plate capacitor, one of them with dielectric and another without dielectric fig(a) shows an usual parallel plate capacitor without dielectric.

Let charge be σ on the plate, with proper sign and A be the area of each plate, E_0 is the electric field without di-electric.

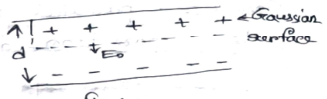


fig:(a)

Therefore, the gauss's law gives,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = \epsilon_0 E_0 A = \sigma$$

$$\text{or, } E_0 = \frac{\sigma}{\epsilon_0 A} \quad \text{---(i)}$$

Now, place the di-electric as shown in fig (b) then the net charge enclosed by gaussian surface is $\sigma - \sigma'$, therefore with dielectric the gauss's law as,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = \sigma - \sigma' \quad \text{---(ii)}$$

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before: 10 $\Rightarrow \epsilon_0 E A = \sigma - \sigma'$

$$\Rightarrow E = \frac{\sigma - \sigma'}{\epsilon_0 A}$$

$$\therefore E = \frac{\sigma}{\epsilon_0 A} - \frac{\sigma'}{\epsilon_0 A} \quad \text{---(iii)}$$

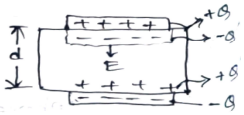


Fig:(b)

On which $-\sigma$ is the induced charge due to polarization but we know dielectric constant

$$K = \frac{E_0}{E}$$

$$\Rightarrow E = \frac{E_0}{K} \quad \text{---(iv)}$$

From equ (i) and (iv), we get.

$$E = \frac{E_0}{K}$$

$$= \frac{\frac{\sigma}{\epsilon_0 A}}{K} \quad \left[\because E_0 = \frac{\sigma}{\epsilon_0 A}\right]$$

$$\therefore E = \frac{\sigma}{K \epsilon_0 A} \quad \text{---(v)}$$

Substitution this value of E in equ (iii), we get,

$$\frac{\sigma}{K \epsilon_0 A} = \frac{\sigma}{\epsilon_0 A} - \frac{\sigma'}{\epsilon_0 A} \quad \text{---(vi)}$$

$$\therefore \sigma - \sigma' = \frac{\sigma}{K} \quad \text{---(vi)}$$

From equ (ii) and (vi), we get,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = \frac{\sigma}{K}$$

$$\therefore \epsilon_0 \oint K \bar{E} \cdot d\bar{s} = \sigma$$

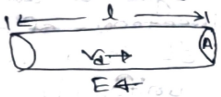
This is the gauss's Law with di-electric.

3. a) Explain the terms current density and drift velocity of electrons inside a conductor.

Answer:

Drift velocity: A potential difference between the two ends of the conductor is established the free electrons drift towards $-E$, that is opposite to the electric field inside the conductor with an average speed. This average speed is known as drift velocity. It is denoted by v_d.

$$\therefore v_d = \frac{J}{ne}$$



Let us consider a conductor of length l and cross-sectional area A.

The electrons move with average velocity v_d towards $-E$. If n is the number of electron per unit volume of the conductor, then the total number of electron inside volume Al is given by,

$$N = nAl \quad \text{---(i)}$$

Total charge flowing inside the conductor

$$q = Ne \quad \text{---(ii)}$$

From (i) and (ii) $\Rightarrow$

$$q = neAl \quad \text{---(iii)}$$

The average time taken by the electrons to travel through the conductor is

$$t = \frac{l}{v_d}$$

We know, $i = \frac{q}{t}$, $\quad J = \frac{I}{A}$

$$\therefore i = \frac{q}{l/v_d}$$

$$= \frac{qv_d}{l}$$

$$= \frac{neAlv_d}{l}$$

$$\therefore i = neAv_d$$

Again, we know, $J = \frac{i}{A}$

$$= \frac{neAv_d}{A}$$

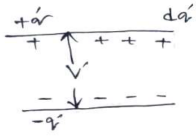
$$\Rightarrow J = nev_d$$

$\therefore$ Drift velocity, $\boxed{v_d = \frac{J}{ne}}$

b) Derive an expression for the energy stored in the electric field of a charged Capacitor.

☞ Energy stored in a charged capacitor.

Or, Proove that, $u = \frac{1}{2}\epsilon_0 E^2$, where symbols are their usual meaning.



Answer:

Let q' be the charge and v' the potential difference established between the plates of the capacitor at any instant during the process of charging. If an additional charge dq' is given to the plates, the work done by the battery is given by $dw = v'dq'$ $\quad \because V' = \frac{q'}{c}$

$$= \frac{q'}{c} dq'$$

Total work done to charge a capacitor to a charge q is

$$w = \int_0^q dw = \int_0^q \frac{q'}{c} dq'$$

$$= \frac{1}{c}\int_0^q q' dq'$$

$$\therefore w = \frac{1}{2}\frac{q^2}{c}$$

This work done is stored as electrostatic potential energy in the capacitor. is

$$U = \frac{1}{2}\frac{q^2}{c}$$

$$= \frac{1}{2}\frac{cV^2}{c}$$

$$\therefore U = \frac{1}{2}cV^2$$

Consider a parallel plate capacitor of area A and plate separation d, The energy of capacitor

$$U = \frac{1}{2}cV^2 = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)\cdot V^2$$

Volume of the space between the plate = Ad.
Energy density u is the potential energy per unit volume,

$$u = \frac{U}{Ad}$$

$$= \frac{\frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)V^2}{Ad}$$

$$= \frac{\epsilon_0 A V^2}{2d} \times \frac{1}{Ad}$$

$$\therefore u = \frac{1}{2}\epsilon_0\left(\frac{V}{d}\right)^2$$

Or, $u = \frac{1}{2}\epsilon_0 E^2$ $\quad \because E = \frac{V}{d}$

Where, u = Energy density.

c) A copper wire of 3m * m ^ 2 Cross-sectional area carries a current of 5A. Find the magnitude of drift velocity for the electrons in the wire.

Example 5.2 A copper wire of 3.0-mm^2 cross-sectional area carries a current of 5.0A. Find the magnitude of the drift velocity for the electrons in the wire.

Soln.

$$j = \frac{I}{A} \qquad A = 3.0\ mm^2 = 3.0 \times 10^{-6} m^2$$

$$= \frac{5.0A}{3.0\times10^{-6} m^2}$$

$$= 1.67 \times 10^6\ A/m^2.$$

The drift velocity is given by

$$V_d = \frac{j}{ne} = \frac{1.67\times10^6\ A.m^2}{n(1.60\times10^{-19} C)} = \frac{1}{n}(1.04 \times 10^{25}\ m^{-2}.s^{-1})$$

where n is the number of charge carriers per unit volume of copper. Assuming one free electron per atom of copper,

$$n = \frac{(\text{Avogardo's number})(\text{density of copper})}{\text{molecular weight of cooper}}$$

$$= \frac{(6.02\times10^{26}\ \text{atoms/kmol})(8920\ kg/m^3)}{63.5\ kg/kmol}$$

$$= 8.5 \times 10^{28}\ atoms/m^3.$$

$$\therefore\ v_d = \frac{1.04\times10^{25}/m^2.s}{8.5\times10^{28}\ atoms/m^3}$$

$$= 0.122 \times 10^{-3}\ m/s$$

$$= 0.122\ mm/s.$$

4. a) State and explain Kirchoff's law.

Or state and explain Kirchhoff's law in a network

1st law or current law:- For any electrical networks the algebraing sum of current meeting a junction is always zero.

i.e, $\Sigma i = 0$.

Explain:- let us consider a junction that $i_1, i_2, i_3, i_4, i_5, i_6$ and i_7 currents are different direction and meeting of a point O. Taking the currents flowing away from the junction as negative and the currents flowing towards the junction as positive.

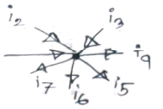


Fig:(a)

we, get,

$$i_1 + i_2 + i_3 + i_4 - i_5 - i_6 - i_7 = 0$$

Or, $i_1 + i_2 + i_3 = i_4 + i_5 + i_6 + i_7$

i.e, incoming current = outgoing current.

Total current entering any junction of a circuit equal the total current leaving that junction.

Kirchhoff's 2nd law/Voltage law:- For a closed circuit, algebraic sum of the products of the current and resistance of the each part of the circuit is equal to the total emfs acting in the circuit.

i.e., $\Sigma iR = \Sigma E$

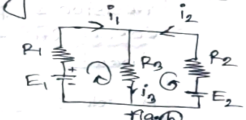


Fig:(b)

Explanation:- For the fig (b), we see that for closed circuit formed by E_1, E_2, R_1, R_2, R_3, we have according to second law,

$$E_1 - E_2 = i_1 R_1 - i_2 R_2$$

and also for the closed circuit formed by E_1, R_1, R_3 and E_1.

We have,

$$E_1 = i_1 R_1 + i_3 R_3$$

Similarly, $E_2 = i_2 R_2 + i_3 R_3$

$\therefore$ For general, $\Sigma E = \Sigma iR$

For a closed circuit the 2nd Law is also called loop theorem.

b) Apply the Kirchoff's law to a wheatstone bridge and find the conditional for bridge balance.

We apply Kirchoff's current law in the shown circuit.

At junction B,

$i_1 = i_g + i_3$

At junction D,

$i_2 + i_g = i_4$

If current through the galvanometer is zero,

$i_g = 0$

thus $i_1 = i_3$

and $i_2 = i_4$

Applying Kirchoff's voltage law for loop ABDA,

$i_1P + i_gG = i_2R$

Applying Kirchoff's voltage law for loop BCDB,

$i_3Q = i_4S + i_gG$

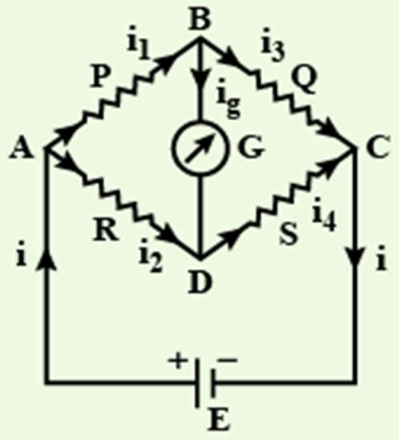
When $i_g = 0$,

$i_1P = i_2R$

and $i_3Q = i_4S$

But $i_1 = i_3$ and $i_2 = i_4$,

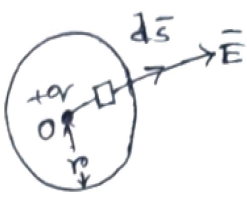
Therefore $\frac{P}{Q} = \frac{R}{S}$



c) Show that Coulomb's law of electrostatics can be deduced from Gauss' law.

Deduction of coulomb's Law from gauss's law:

Let us apply gauss's law to an isolated point charge +q lying in vacuum, the gaussian surface is spherical surface of radius r having its centre at the point charge $\bar{E}$ and $d\bar{s}$ at any point on the surface are directed radially outward angle between $\bar{E}$ and $d\bar{s}$ is zero.



$$\therefore \bar{E}\cdot d\bar{s} = E\cdot ds\cos 0^\circ$$

$$= E\,ds$$

Applying gauss's law,

$$\oint \bar{E}\cdot d\bar{s} = \oint E\,ds = \frac{q}{\varepsilon_0}$$

E is constant at all point on the surface,

Thus, $$E\oint ds = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E\cdot(4\pi r^2) = \frac{q}{\varepsilon_0} \qquad \left[\because \oint d\bar{s} = 4\pi r^2\right]$$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{r^2}$$

Let us put a second point charge q' on this surface, the force $\bar{F}$ that acts on this charge q' is $F = q'E$

$$= \frac{1}{4\pi\varepsilon_0}\cdot\frac{qq'}{r^2}$$